SAT Memo 02 — Formal Analysis of Your Solution

VCE Algorithmics (HESS) · Unit 4 Outcome 1 · SAT Criteria 5–7

*Issued Week 4 · **Due Tuesday 18 August 2026** (Submission 2), with Amendments A03 & A04*

The engineering situation

Head Office has approved your Operation facility design from Memo 01. Before committing hardware, the engineering board requires a **performance audit**: a formal analysis of how your routing solution's running time scales as facilities grow. Your audit will determine whether the current design ships as-is, or which component is redesigned in the next phase.

What you must deliver

1. **Memo 02** — a formal time complexity analysis of your Memo 01 data model and algorithm combination, comprising:
    - a **module inventory**: every module analysed with its loop/recursion structure, the analysis tool used (loop counting, amortised edge counting, call graph, recurrence + expansion, or Master Theorem where the form applies), the tightest worst-case bound, and a one-sentence justification; **C5**
    - a **tally**: module bounds composed along the control flow to a justified headline bound for the whole solution — follow the VCAA FriendshipRecommender exemplar's format; **C5**
    - a **consequences section**: what the headline bound means for the real facility — realistic input sizes, projected cost at those sizes and at growth, and the design decision that follows; **C6**
    - a **comparison section**: your chosen algorithm(s) against at least one credible alternative for the same task, with both bounds, the conditions under which each wins, and a conclusion. **C7**
2. **Completed workbook** (`marimo_memo02`) — your working: inventory table, scaling projections, and paragraph drafts.
3. **Amendment A03** and **Amendment A04** (below).

Amendments

*As with A01/A02, amendments are Head Office change requests applied to **your** facility.*

A03 — The night-shift audit **C7**

Overnight, the facility runs a reduced service: routes are needed from the **single** security office to all rooms, recomputed whenever a corridor closes. Your Memo 01 solution computes more than this requires. Specify the cheapest studied algorithm that meets the night-shift requirement, state its worst-case bound on your facility graph, and compare it against running your existing solution — including the crossover conditions under which your original approach would still be preferable.

A04 — The expansion stress test **C6**

Head Office proposes federating ten facilities into one navigation system (roughly 10× your current $|V|$, with sparse links between facilities). Using your headline bound, project the cost of your current solution at the federated scale, state whether it remains feasible for (a) one-off precomputation and (b) per-closure recomputation, and identify which module becomes the bottleneck. *Do not redesign yet — that is Unit 4 Outcome 2 (Memo 03).*

What is assessed

C5	Skills in determining the time complexity of algorithms.
C6	Understanding of the consequences of an algorithm's time complexity on its real-world application.
C7	Skills in the comparison of the time complexities of algorithmic solutions to a real-world / applied problem.

Note: the in-class **Test 1** (24 July) also assessed C5. Submission 2 combines your test result with this memo across C5–C7.

Timeline and support

Week 4, P3	Workshop 1 — module inventory and first bounds (in class)
Week 5, P2	Workshop 2 — consequences, comparisons, A03/A04 (in class)
Mon 17 Aug	Checkpoint — workbook complete, declaration signed
Tue 18 Aug	**Submission 2 due**

Authentication requirements

- Your facility **seed must match** Submissions 1 and 2 — it identifies your unique facility and appears on your cover sheet.
- Substantive work must be done in the class workshops and in your own workbook. Save your workbook at each checkpoint.
- You may discuss *methods* with other students; the analysis of *your* facility must be your own. Cite any external resources.
- The VCAA authentication record form must be signed at each checkpoint.

Advice from the marking exemplar

The VCAA exemplar (FriendshipRecommender) analyses each module in isolation, states the tool, then tallies: *"$2 \cdot O(|V|) + O(|V|^2) + \ldots + O(|V|^3) \quad O(|V|^3)$, since the Floyd–Warshall contribution is the highest order polynomial."* That sentence shape — module bounds, then the dominant term with a reason — is the standard your tally is marked against. Bounds without justification score at most half; consequences that merely restate the bound ("it is slow") do not meet C6.